Hypernatremia is defined as serum sodium greater than 145 mEq/L and indicates either an absolute or relative water deficit. Normally, an elevated sodium level stimulates thirst and antidiuretic hormone secretion. Because increased ingestion of water relieves hypernatremia, it usually only develops in patients unable to ingest fluids such as obtunded patients, anesthetized patients, or infants.

Symptoms of hypernatremia include stupor, coma, seizures, hypovolemia, renal insufficiency, and decreased ability to concentrate urine. Hypernatremia may also develop as a result of diabetes insipidus.

Treating hypernatremia should be done slowly as rapid restoration of the sodium level to normal may precipitate seizures or cerebral edema. The total body water deficit should be replace over 24 to 48 hours.

Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1406.

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